eBook

# Streamlining Supply Chains With Data and AI

databricks

SAP

# Contents

Executive Summary ..... 5
Embracing the Data-Driven Supply Chain ..... 7
Applying Data and AI Across Industry Use Cases ..... 8
Challenges in Harnessing Data and AI ..... 14
Why Industries Are Choosing Databricks ..... 18
Simplify Building Data and AI Applications With Your SAP and Enterprise Data ..... 27
Next Steps to Data and AI-Supported Inventory ..... 30
About Databricks ..... 31

# Executive Summary

Across today's dynamic industries — from Retail and Consumer Goods (RCG) to Manufacturing and Energy as well as Healthcare and Life Sciences (HLS) — effective inventory and supply chain management remains one of the most pressing hurdles to operational efficiency, business resilience and customer satisfaction. Organizations must balance demand forecasting, production planning, logistics coordination and real-time decision-making across global, interconnected supply chains. The complexity of coordinating these moving parts becomes even more pronounced when external disruptions — such as geopolitical instability, climate events and rapid market shifts — introduce volatility and unpredictability that heighten the urgency for smarter, more responsive operations.

Modern enterprises require end-to-end visibility and control over every stage of the inventory lifecycle — from raw materials and work-in-progress to finished goods and distribution — while meeting growing expectations for speed, transparency and sustainability. Each stage brings its drivers, analytics needs and decision-making timelines, yet success hinges on how well these phases function in unison. These needs are further amplified in regulated environments like Healthcare, high-volume sectors like Retail, and capital-intensive operations like Energy and Manufacturing. This complexity only grows as supply chains stretch across continents and involve numerous partners, often requiring decisions to be made in seconds — yet planning takes months. Achieving true optimization calls for synchronizing supply and demand with the help of real-time insights, predictive intelligence and AI-driven automation.

These types of data-rich industries increasingly seek a unified, real-time view of their supply chains through AI-driven visibility and early disruption detection. But data engineering challenges often stand in the way: critical information is fragmented across multiple partners, requiring time-consuming manual extraction and integration — if it's even accessible at all. Historically, supply chain solutions were pieced together from disconnected systems, resulting in lags and inefficiencies. Modern businesses, however, need to incorporate both internal and external data — ranging from seasonal and market conditions to unstructured sources like satellite imagery and social media — to gain a holistic operational perspective.

According to MIT Technology Review, AI adoption falters primarily due to subpar data quality (57%), insufficient data integration (54%) and weak governance (47%). These issues impede the development of robust data pipelines and limit the scalability of AI. Compounding these hurdles is a widespread distrust in AI models, which often lack transparency and can be difficult to validate, especially in safety-critical environments. Security concerns, including data privacy and vulnerability to cyberattacks, further erode confidence. To fully realize AI-driven supply chain excellence and resilience, enterprises must tackle data engineering at its core — centralizing data on a single platform, ensuring reliable pipelines and embedding governance to enable the speed, visibility and trust required for transformative insights.

Enterprises across the supply chain face mounting pressure to unify data, reduce risk and lower costs. The Databricks Data Intelligence Platform addresses these needs in three critical ways: first, its best-in-class engineering capabilities simplify real-time data integration from diverse sources; second, its native governance features instill confidence in data quality; and third, its scalable AI delivers accurate, cost-effective insights. By centralizing data and analytics on a single platform, organizations gain the agility to respond swiftly to external pressures, minimize waste, optimize inventory and free up working capital. In today's fast-paced market, a modern, data-driven supply chain is no longer optional — it's a must-have for sustainable, profitable growth.

> "Annual global AI software revenue [which can be applied to inventory management in consumer goods] is forecast to grow from $10.1B in 2018 to $126B by 2025."
>
> — Omdia

# Embracing the Data-Driven Supply Chain

Across industries, data has rapidly become one of the most valuable strategic assets. As global supply chains grow more complex and consumer expectations rise, organizations are turning to artificial intelligence (AI) and machine learning (ML) to extract actionable intelligence from vast amounts of data — not just to keep pace but to lead in their respective markets.

The foundation of this transformation lies in the convergence of cloud computing, Internet of Things (IoT) devices and real-time analytics, which collectively enable continuous visibility, faster decision-making and more responsive operations. In RCG, companies use AI to anticipate demand shifts, personalize customer experiences and streamline fulfillment. Manufacturers leverage digital twins and predictive maintenance models powered by IoT data to reduce downtime and optimize asset utilization. In HLS, real-time analytics help manage clinical supply chains, ensure the availability of critical treatments, and accelerate research and development through smarter data use.

Adoption is gaining speed because these technologies don't just solve isolated problems; they unlock entirely new ways of working. By consolidating data across siloed systems and making it accessible through scalable, cloud-native platforms, enterprises can deploy predictive models, automation and AI agents that operate at scale, enabling intelligent workflows and proactive supply chain adjustments.



# Applying Data and AI Across Industry Use Cases

## Improving real-time, end-to-end visibility across the value chain

Achieving full visibility across today's global supply chains requires more than traditional supply and demand planning. According to leading consultancies like McKinsey, Deloitte and BCG, organizations must now incorporate external market signals — such as geopolitical events, environmental conditions and emerging health risks — well in advance to proactively address potential crises. Early detection of these disruptions is paramount; waiting until a concern becomes an emergency often leads to higher costs and more severe operational impacts.

By unifying vast and varied data sources on the Databricks Data Intelligence Platform, companies can capture and analyze real-time signals spanning everything from tariff changes and civil strife to droughts, wildfires and epidemics. Integrating structured and unstructured data — including IoT sensors, cloud applications, enterprise resource planning (ERP) systems, satellite imagery and social media — through the Databricks Marketplace powers real-time analytics and predictive models with thousands of supplemental datasets that enrich internal data. Meanwhile, Databricks Delta Sharing significantly lowers the economic and technical barriers to ingesting these signals, delivering up-to-the-minute insights without burdensome manual integration.

Machine learning and predictive analytics within Databricks then convert these signals into actionable intelligence. Instead of merely responding to disruptions as they occur, businesses can preemptively adjust production schedules, reroute logistics or diversify suppliers. Digital twins simulate the impact of potential changes before action is taken, while AI agents and automation orchestrate responses across systems without human delay. This level of proactive planning not only mitigates risk but also strengthens relationships with suppliers and customers alike. By harnessing AI to illuminate blind spots and gain real-time data across the value chain, enterprises create a resilient, agile supply network equipped to thrive in an increasingly unpredictable world.

## CHOOSING THE BEST MATERIALS, VENDORS AND SOURCES

Securing high-quality, reliable and cost-effective inputs remains crucial for companies; however, supply chain disruptions commonly begin with third-party vendors, particularly among second- and third-tier suppliers, whose operations are less visible. Leading consultancies emphasize the importance of proactive sourcing strategies that factor in vendor performance, on-time delivery history and risk metrics. By consolidating internal data with external signals through the Databricks Data Intelligence Platform, procurement teams gain a holistic, AI-driven view of supplier reliability and cost, as well as geopolitical, environmental and economic indicators that influence stability.

With real-time insights at their disposal, organizations can quickly assess and rank multiple suppliers — especially those at deeper tiers — based on on-time-delivery records and other performance criteria. This empowers teams to respond rapidly to disruptions or market shifts by pivoting to alternative sources before minor issues escalate into full-blown crises. Predictive analytics further refines procurement decisions by anticipating demand spikes or raw material shortages, minimizing stockouts and excess inventory. Automation and AI agents allow faster reallocation of resources or onboarding of alternative vendors when performance dips or threats emerge.

Technologies like robotics, IoT and digital twins further enhance sourcing by connecting procurement with real-time production and operating conditions, helping to optimize not just whom to source from, but when and how. Meanwhile, Delta Sharing reduces the overhead of integrating data from outside partners, delivering timely intelligence without the bottlenecks of manual processes. Ultimately, this centralized, AI-enabled approach fosters a more resilient supply network, ensuring organizations can adapt swiftly and accurately to unforeseen challenges across their global value chains.

## Optimizing supply and demand with precision

Managing inventory involves multiple stages — from raw materials to work-in-progress, finished goods, distribution centers and retail shelves — each with its own data needs and decision-making complexities. Effective planning at every stage requires advanced analytics and AI to handle fluctuating demand signals, supply constraints and logistics challenges across a range of areas. The following are the primary decision areas companies must master to optimize inventory and balance cost with customer satisfaction:

**Demand forecasting and planning** encompass both near-term and long-term needs across a wide range of products, including managing variability from new product introductions, seasonal fluctuations and industry trends. Capturing and interpreting diverse demand signals — including point-of-sale systems, IoT data, cloud technology, social media sentiment, historical usage patterns and promotional activities — remains a primary challenge. Launches for new products and seasonal peaks compound the difficulty, as historical data for these scenarios may be scarce or less reliable. Nonetheless, accurate forecasting is critical for preventing shortages, minimizing excess inventory, controlling costs and improving service levels across the supply chain.

**Supply forecasting and capacity management** involve aligning demand projections with supplier lead times, production schedules and real-time capacity constraints. However, many organizations struggle to track the availability of raw materials, particularly when suppliers are distributed globally or span multiple tiers, which can lead to hidden risks in both cost and continuity. Proactive responses to potential disruptions — such as labor strikes, shortages or geographic events — rely on real-time signals from suppliers and early-warning indicators like weather and geopolitical data. When supply and demand forecasts are properly synchronized, operations teams can schedule production efficiently, optimize resource utilization and minimize bottlenecks, ultimately boosting overall operational effectiveness.

**Production planning** involves determining optimal batch sizes, sequencing and run times for each facility. Volatility in both demand and variations in material availability make it challenging to schedule production runs without leaving equipment idle or creating surplus inventory. Advanced optimization models, which combine AI and scenario planning, are increasingly crucial for balancing production efficiency with the flexibility to adapt to external events. By streamlining production planning, organizations can reduce costs associated with downtime or overproduction while remaining highly responsive to changing market demands, regulatory requirements and patient and customer needs.

**Logistics optimization** focuses on ensuring the timely, cost-effective movement of goods, materials or equipment across increasingly complex supply networks. However, route planning can quickly become challenging when dealing with multiple modes of transportation, various distribution hubs and geographies, and fluctuating fuel or capacity constraints. Unforeseen events — such as extreme weather, route closures, infrastructure issues or geopolitical events — can then trigger cascading delays that ripple throughout the supply chain. By leveraging AI-driven route optimization and real-time supply chain visibility, organizations can mitigate these disruptions, minimize delays, lower transportation costs, and ensure the availability of critical products or resources where and when they're needed most.

**Inventory allocation and SKU rationalization** focus on determining what to stock, where to stock it and in what quantities, to meet service levels while controlling costs. Organizations must accurately forecast consumer demand for both high-volume and niche products or services, and align those forecasts with real-time data on inventory, supply availability and production or distribution capacity. Balancing a broad product or asset portfolio against profitability and optimal inventory turns is particularly challenging when multiple regions or channels are involved. By leveraging AI-driven allocation models and rationalization initiatives, businesses can ensure that the right materials, components or products are available in the right locations at the right time, reducing shortages and minimizing excess inventory and overall total cost of ownership (TCO). Additionally, AI can support ongoing portfolio optimization by identifying low-performing or redundant items and modeling the operational and financial impact of potential changes.

**Product availability and retail execution** at the point of delivery focus on ensuring that finished goods, components or critical supplies are consistently in stock and available for end users. A lack of real-time visibility into upstream inventory and distribution status can create mismatches between supply and actual consumer demand. Additionally, unanticipated demand spikes, such as those driven by promotional campaigns, clinical needs or operational surges, can quickly deplete stock if not factored into replenishment strategies. By leveraging real-time replenishment and predictive analytics, organizations can improve availability, increase stakeholder satisfaction and reduce the risk of lost revenue or operational delays.

## WHY DO THESE BARRIERS REMAIN?

Data fragmentation remains a significant obstacle to effective supply chain oversight. Incomplete visibility into supplier performance — especially among lower-tier vendors — and inconsistent tracking of production capacity can undermine even the most precise demand forecasts. Four key factors tend to exacerbate these challenges: limited access to high-quality, governed data; data that is either outdated or arrives too slowly to guide timely decisions; analytics systems that cannot handle diverse modeling techniques and large, granular datasets within business SLAs; and difficult-to-use tools that discourage end-user adoption. These issues are exacerbated by constantly shifting consumer preferences and global trade conditions, making it more challenging to scale AI solutions that rely on accurate, up-to-date data.

The Databricks Data Intelligence Platform tackles these hurdles by centralizing data from all stages of the supply chain and enabling real-time ingestion from virtually any source. Databricks Unity Catalog enforces data quality standards, ensuring that AI models and analytics engines — capable of handling even the most demanding workloads — are fueled with reliable information. Business-friendly tools like AI/BI Genie facilitate user adoption, allowing nontechnical stakeholders to glean insights with minimal friction. Finally, Delta Sharing allows organizations to distribute these insights in real time to partners, bypassing costly integrations and IT overhead. By unifying data, enforcing governance and simplifying both AI workloads and data sharing, this approach supports more accurate demand forecasting, improved production scheduling and portfolio optimization — ultimately ensuring availability where it's needed, when it's needed.

## Delivering better data collaboration across the supply chain

Inefficient collaboration across supply chain partners leads to trillions of dollars in lost opportunities throughout the supply chain. Beyond the direct loss in revenue, these inefficiencies also hinder profitability by driving up expedited shipping costs while further damaging brand sentiment, customer loyalty and service level expectations. These issues are symptoms of a broader problem: poor supply chain visibility and inefficient collaboration among manufacturers, distributors, service providers and end users.

**Increasing visibility into real-time data signals and improving prediction accuracy are key to addressing this problem. However, the core barrier remains: most organizations still lack scalable, cost-effective ways to collaborate with a broad range of partners.**

Traditional data-sharing methods are often costly, complex and poorly aligned with the needs of today's fast-moving supply chains. Legacy tools require multiple costly systems to exchange data, while newer solutions are dependent on expensive, proprietary technologies. As a result, many companies collaborate with only their largest partners, leaving significant untapped potential for improvement. Delays in exchanging critical information further compound data sharing challenges.

Delta Sharing allows suppliers to share data in real time with partners across different technologies and clouds, eliminating both licensing fees and technical barriers. Freed from these constraints, suppliers can broaden their collaborative networks, resulting in greater efficiency and reduced costs for data management and exchange. By prioritizing seamless sharing, organizations also enhance their responsiveness to shifting demands, streamline operations and accelerate innovation, particularly by easing experimentation with AI and other emerging technologies.

> "Companies in the telecommunications, electric power, natural gas and healthcare industries have found that AI forecasting engines can automate up to 50% of workforce-management tasks, leading to cost reductions of 10 to 15% while gradually improving hiring decisions and operational resilience."
>
> — McKinsey

# Challenges in Harnessing Data and AI

## Channel and distribution complexity

While expanding distribution channels through e-commerce and direct-to-consumer (DTC) channels provides more revenue opportunities, it also complicates inventory management due to increased complexity and variability in demand. Across industries, each channel has its unique demand patterns, fulfillment expectations, and operational requirements. Traditional outlets, digital marketplaces, institutional buyers and end-user delivery all vary in velocity, volume, and consumption behavior, forcing organizations to manage multiple inventory pools with different turnover rates and stocking strategies.

For example, while bulk shipments may suit traditional distribution models, DTC and just-in-time delivery approaches often require smaller, more frequent orders, particularly in healthcare settings or decentralized energy deployments. To meet modern expectations for fast and accurate fulfillment, enterprises need real-time visibility and coordination across all channels. To maintain responsiveness and ensure product or asset availability where and when it's needed, organizations must invest in advanced analytics and AI solutions to synchronize inventory, forecast demand and optimize the value chain across fragmented and dynamic distribution landscapes.

## Supply chain volatility and globalization

Global supply chain disruptions can significantly upend inventory management, supply continuity and end-user demand across industries. Events like geopolitical tensions, natural disasters, transportation bottlenecks and fluctuating raw material prices can cause sudden and unpredictable delays, shortages and cost spikes. The COVID-19 pandemic highlighted the fragility of global supply networks, underscoring the need for enterprises to adapt quickly to maintain inventory stability and meet demand. Adapting to these unknowns is critical for organizations to maintain optimal inventory levels. However, supply chain volatility often forces businesses to revise inventory strategies on short notice due to changes in supplier reliability, logistics constraints or manufacturing capacity. The result is frequently a costly imbalance — either excess inventory that strains working capital or stockouts that disrupt service and customer satisfaction.

Globalization further exacerbates these challenges, expanding the intricacies of supply networks and their geographical reach. Managing supply chains across multiple regions involves navigating different regulatory environments, fluctuations in frequency and cultural differences. Extended lead times and increased risk of disruptions in global supply chains necessitate a more sophisticated approach to inventory management. Globalization also increases the interdependencies within the value chain. A disruption in one part of the world can ripple across the entire network. This is especially critical in sectors like healthcare, where delivery delays can directly impact patient outcomes, or in Energy, where supply interruptions can stall infrastructure projects or operations.

As a result, many organizations are exploring more localized or regional approaches to production and sourcing. While this shift can enhance resilience, it also introduces new layers of complexity for inventory management and distribution. To navigate this environment effectively, companies need to invest in advanced data intelligence, real-time visibility and predictive analytics to detect potential issues early and optimize inventory strategies across their global operations.

## Sustainability and ethical sourcing

Across industries, from CPG to Manufacturing and Energy, as well as healthcare and life sciences, organizations face growing pressure to prioritize sustainability, ethical sourcing and supply chain transparency. Customers, patients, regulators and investors increasingly demand products and services that are responsibly sourced, environmentally conscious and socially ethical. This shift forces enterprises to vet and select suppliers based on sustainability credentials, such as certifications for renewable sourcing, low carbon emissions or fair labor practices, which can limit choices and drive up expenses. While these commitments support long-term resilience and brand trust, they often reduce supplier flexibility, increase costs and introduce added complexity into the supply chain.

Regulatory pressure further intensifies these challenges. Governments and international bodies are enacting stricter rules around carbon neutrality, traceability, responsible sourcing and environmental impact disclosures. Compliance with these regulations often requires additional documentation and verification processes, adding to the time and resources needed for operational management. These constraints make it more difficult for CPG companies to maintain a consistent supply of materials and products. In sectors such as Healthcare and Energy, where material sourcing is already subject to compliance and safety standards, these additional sustainability requirements pose a new challenge. Noncompliance often results in significant financial penalties, business disruptions and damage to the organization's reputation. Balancing these demands while maintaining efficient inventory management requires sophisticated tracking systems and robust supplier relationships, adding new variables to supply chain operations.

> "Products making ESG [environmental, social and governance] related claims averaged 28% cumulative growth over the past five-year period, versus 20% for products that made no such claims."
>
> — McKinsey

## Applying advanced technologies

Game-changing technologies, such as AI, ML, real-time data processing and data sharing, present both opportunities and challenges for inventory and supply chain management across various industries. While these technologies offer powerful tools for predicting demand and optimizing stock levels, they also require significant investments in data infrastructure and governance, as well as upskilling or hiring capable team members and integration with existing systems. The rise of autonomous logistics, for example — including self-optimizing transportation networks, automated warehouses and AI-driven fulfillment systems — is fundamentally reshaping how organizations forecast demand and plan supply chain operations. Autonomous systems generate massive volumes of real-time operational data, from fleet performance and inventory movement to route optimization and last-mile delivery. When these systems are integrated with demand planning systems, enterprises gain more accurate forecasting by aligning supply chain actions with actual market behavior, rather than solely historical trends.

While it's essential to adopt and leverage innovation, implementing AI and ML strategies with these new advances requires vast amounts of structured and unstructured data, necessitating robust and secure data management practices, as well as advanced analytics capabilities. This shift often leads to increased operational costs and a steep learning curve for data practitioners, complicating transition and management processes throughout the supply chain.

For real-time data processing and data sharing, enterprises need seamless coordination across multiple stakeholders and technology partners within the supply chain. Maintaining data governance and security across various platforms and partners is critical yet challenging when applying these tools. The need for continuous monitoring and instant decision-making can strain resources and systems, making it difficult to maintain consistent inventory levels. Additionally, reliance on sophisticated technologies can expose companies to vulnerabilities like system failures and cyberattacks, further complicating inventory management and data reliability. Balancing the benefits of advanced technologies with these new challenges requires a strategic and well-coordinated approach to data governance.

# Why Industries Are Choosing Databricks

The Databricks Data Intelligence Platform is the industry-leading solution for data and AI, trusted across a wide range of sectors. As organizations embrace data-driven innovation, Databricks helps shape market trends while minimizing the risks of adopting emerging technologies. With powerful, differentiated capabilities, the platform enables transformative use cases across the enterprise — from customer intelligence and fraud detection to supply chain optimization — unlocking deeper insights and operational efficiency at scale.

> "Over 60% of the Fortune 500 rely on Databricks."
>
> — Databricks

## Databricks differentiator #1: Real-time data warehousing

In industries where timing, precision and responsiveness are crucial, inventory management must be intelligent and immediate to maintain an adaptive and resilient supply chain. The Databricks Data Intelligence Platform instantly enables data integration from dozens of internal and external sources, providing real-time data warehousing that facilitates faster decision-making and more agile operations.

With real-time data warehousing, businesses can eliminate delays that typically hinder inventory optimization. In RCG, this means dynamically adjusting stock levels in response to shifting consumer demand or promotion-driven surges. In Manufacturing and Energy, predictive analytics and digital twins simulate supply and production scenarios to align inventory with plant operations or field requirements, ensuring optimal utilization of resources.

Within HLS, where availability can be a matter of patient safety, the platform supports automated replenishment strategies and demand sensing to ensure critical medical supplies and treatments are where they're needed most. By unifying data across ERP systems, CRMs, and first- and third-party data, data products can be made available to the data warehouse across customers, contracts, invoices, orders, suppliers, formularies, distribution centers and beyond. This results in users asking questions of their data in natural language, like "Which shortages will have the most impact on my customer experience?" or "How can I mitigate risk to product availability?" and even "Which products are most likely to face shortages within five days?"

The Databricks Platform transforms inventory management from a reactive process into a predictive, resilient and continuously optimized engine of competitive advantage.

- **Delta Lake** has Auto Loader technology to support massive data ingestion in near real-time. Organizations are limited only by the frequency at which source systems can supply data, providing a continuous flow of up-to-date information crucial for making timely and intelligent decisions.
- **Lakeflow** enhances data operations by unifying ingestion from CRM, SAP, file systems and databases via Lakeflow Connect, while Lakeflow Declarative Pipelines automate robust batch and streaming transformations. Paired with highly reliable orchestration through Lakeflow Jobs, Lakeflow enables real-time insights to streamline supply chain operations.
- **Business intelligence** improves every part of the supply chain by turning real-time data into actionable insights. Databricks AI/BI understands your enterprise data and business context, securely delivering accurate answers to natural language questions tailored to your organization and supply chain needs.

**Maintain optimal inventory levels**

**Efficiently reduce waste**

**Respond quickly to market changes**

With Databricks, Walgreens built a smart data ecosystem capable of forecasting medication demand and reallocating inventory across its network of nearly 9,000 pharmacies. The platform also delivers near real-time insights and personalized recommendations to support pharmacists in offering tailored care to patients.

> "With Databricks, we can create insights from data to better understand how best to engage and communicate with patients, from sharing vital information about medications to guiding on their journey toward a healthier life."
>
> — Vice President, Pharmacy and Healthcare Platform Technology, Walgreens

## Databricks differentiator #2: Data governance with Unity Catalog

Effective inventory and demand management depend on the availability of clean, reliable data sourced from multiple internal and external systems, including ERP systems, IoT devices, supplier feeds and patient data streams. Databricks Unity Catalog is a centralized governance layer for managing data quality and AI within the Databricks Platform, delivering the following capabilities:

- **Comprehensive data monitoring**, which allows administrators to oversee data integrity and reliability across the organization. If issues arise, admins can review changes in the data lineage to identify problems and perform "time travel" to return to previous data states. By consolidating data, organizations can maintain more efficient and consistent inventory records, which are crucial for predictive analytics, AI-driven forecasting and real-time decision-making within dynamic environments such as omnichannel retail, decentralized energy networks and clinical supply chains.
- **Unstructured data management**, a necessary and ongoing task for powering AI applications because 80% of AI success depends on data quality. Govern structured and unstructured data, ML models, notebooks, dashboards and files on any cloud or platform to streamline data processing with reliability for broader and more advanced applications. This unified oversight enables broader applications, such as digital twins for supply chain simulation, AI agents for automated inventory control and robotics coordination in warehouse operations, all while maintaining data compliance and trust.
- **Automated administration** using AI-driven tasks that reduce technical debt and improve productivity to lower TCO. Features like liquid clustering automatically optimize partitioning and indices to improve query speed, and predictive I/O uses access patterns to support query performance for real-time restocking, just-in-time delivery or urgent demand spikes.
- **Unobstructed discoverability** for data scientists, analysts and engineers through a single pane of glass that accelerates time to insights by eliminating time-consuming searches across disparate data sources. This unified data access streamlines the development of advanced analytics and AI applications, including cybersecurity threat detection, intelligent procurement and automated anomaly detection across supply chain nodes.

**Simplify regulatory compliance** + **Boost data team productivity** + **Confidently trust your data**

Bayer implemented Databricks with Unity Catalog to create a unified and secure environment for data governance and analytics across its global operations. With controlled access and meticulously tracked data lineage, Bayer is consolidating different data sources from various data providers to harmonize the data for reuse. Now, with only one endpoint, authorized business users can analyze data across data products and coordinate assets to select their desired data and build meaningful reports.

> "With the Databricks Data Intelligence Platform and Unity Catalog, we are currently enabling our business users to analyze all data for which they have authorizations with just one single endpoint. So, they can easily scroll across all data products and coordinated assets to select the data they need and build the reports that matter to them."
>
> — André Wuthenow, Lead Data Architect, Bayer

## Databricks differentiator #3: Scalable AI and intelligence agents

In the past, companies had to compromise on what they could analyze due to constraints in the scalability of predictive systems and narrow processing windows to meet business service level agreements (SLAs). The Databricks Data Intelligence Platform eliminates these constraints, allowing for advanced methods of analysis and synthesis through generative AI.

- **Databricks ML supports extensive prediction scaling, enabling companies to perform millions of predictions within tight service windows.** This capability allows for advanced applications like real-time demand forecasting, predictive maintenance, digital twin simulations and clinical trial modeling, serving as the foundation for agentic AI systems that act on these insights in real time. By replacing performance bottlenecks with the agility and speed required to respond to market changes swiftly and purposefully, organizations can move from reactive analysis to proactive, autonomous action.
- **Mosaic AI enables the creation of generative and agentic AI solutions** with meticulous control over data quality and output, providing a highly efficient cost structure for training and serving AI models for use cases like AI agents for autonomous inventory management, automated documentation in healthcare and customer personalization in retail. The efficiency translates to significant savings, allowing organizations to operate at 50% to 67% less cost than with other systems.

  With data remaining on the centralized Databricks Platform, companies maintain end-to-end data governance, reinforcing security and regulatory compliance, particularly in regulated industries like HLS and Energy. This approach safeguards sensitive information and leverages the full potential of in-house data resources. Companies have complete control over their AI models — commercial or open source — which provides flexibility for rapid adjustments and fine-tuning as needed.
- **Data rooms empower frontline employees to query company data** using the enterprise's specific language and terminology to foster a data-driven culture across the organization. With governed data access, nontechnical users can interact with AI agents to quickly and independently obtain answers without specialized knowledge.

**Enhance efficiency to lower AI costs**

**Democratize data use through AI agents**

**Optimize data-based decision-making**

Shell selected Databricks as a core component of its Shell.ai platform to overcome data complexity and scale its AI-driven energy initiatives. With Databricks, Shell empowers hundreds of data engineers, scientists and analysts to collaborate more effectively, accelerating innovation in areas like operational efficiency, customer engagement and the transition to cleaner, renewable energy.

> "The usage of Databricks over the years has broadened significantly. We started out using Databricks as a big data and AI platform, but the scope has broadened. We have an entirely different class of citizen engineers and data scientists who are using it as a modern business intelligence tool to make smarter business decisions."
>
> — Daniel Jeavons, Vice President, Computational Science and Digital Innovation, Shell

## Databricks differentiator #4: Secure collaboration with Delta Sharing

High-quality business decisions are intrinsically linked to the freshness and accuracy of the underlying data. Data's relevance and usefulness diminish as data ages, leading to potential misinformed choices. Delta Sharing natively integrates with Unity Catalog, inviting open, cross-platform data sharing using zero-copy, zero ETL integration with SaaS applications, partners and marketplace providers.

With Delta Sharing, enterprises can share data across different systems without the traditional bottlenecks associated with data copying and transformation processes. Companies get a unified view of inventory and other critical data points across the entire value chain, derived from dozens of internal and external systems. This capability fulfills use cases like predictive analytics for demand planning, digital twin simulations for production and logistics, and AI agents that trigger automated responses based on live operational data.

- **Instant data availability** in transactional systems provides decision-makers with real-time data updates for quick responses to market changes, inventory fluctuations and consumer demands. Leveraging current information in real time helps companies manage inventory with less waste and increased supply chain efficiency. In RCG, organizations gain a consolidated view of the value chain, enabling them to optimize performance across all channels. In manufacturing, near-instant updates enable automated reordering, robotic coordination and more agile production cycles. For HLS, clinical supply chains and regulatory reporting benefit from secure, real-time access to inventory, research and treatment data across facilities.
- **Low-cost data integration** removes infrastructure, monitoring and maintenance expenses to give enterprises a streamlined integration approach without expensive platforms and ongoing management. Financial efficiency lets organizations shift investment away from maintaining integration infrastructure and toward strategic initiatives like cybersecurity, AI scalability or expanding analytics capabilities to fuel smarter growth.

**Gain end-to-end visibility and agility**

**Plan and forecast with supply chain partners**

**Manage, govern, audit and track usage**

Moving to the Databricks Data Intelligence Platform enabled Reckitt to empower its employees by democratizing data through the lakehouse architecture, allowing its over 7,000 users to unlock new data perspectives and better predict market volatility.

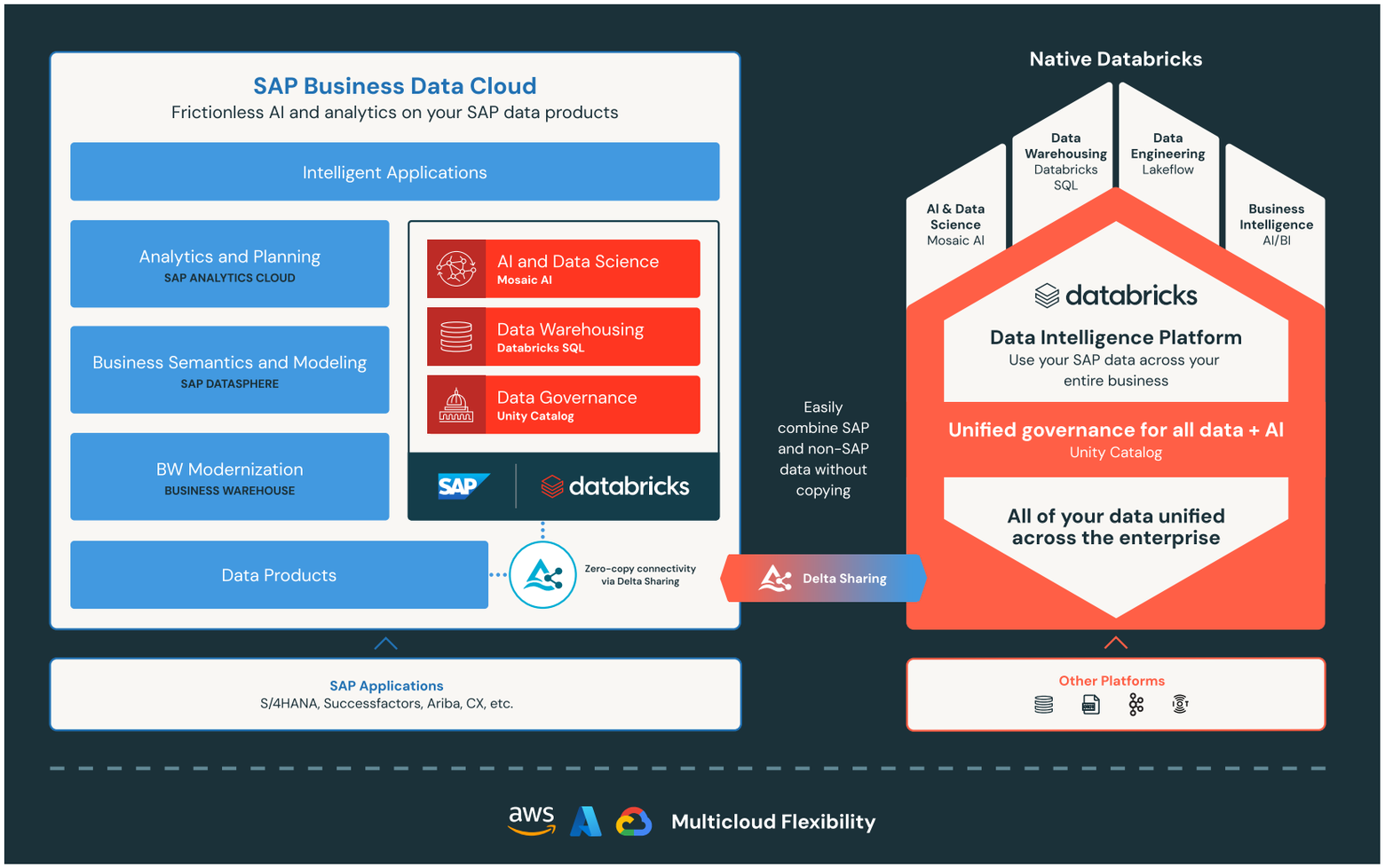
"Databricks enables us to outperform and deliver best-in-class customer experiences in every market. With its Data Intelligence Platform, our stakeholders and developers can work more collaboratively and unlock critical trends and insights needed to serve our 180 million customers."

— Sergiy Tkachuk, Head of Data Science, Reckitt

# Simplify Building Data and AI Applications With Your SAP and Enterprise Data

Every enterprise wants to improve its supply chains by extracting more value from SAP data, particularly by combining it with data from other systems. However, those systems are varied, and many still sit on-premises in legacy platforms, making it challenging and expensive to blend with external datasets. Custom ETL pipelines and data replication further drive up costs and compromise reliability, agility and trust in data — key ingredients for AI-driven transformation.

## Databricks and SAP: Domain-specific AI for your supply chain



SAP and Databricks now deliver an integrated, cloud-native solution. This partnership brings together SAP's robust data, analytics and planning capabilities, via SAP Business Data Cloud, and Databricks' industry-leading AI/ML, data warehousing, data engineering and data governance capabilities — enabling transformative insights with simplicity and speed.

SAP and Databricks provide a zero-copy, live data access approach to SAP data from Databricks — via Delta Sharing — that eliminates the need for data integration and replication. Organizations can access all of their SAP data and join it with external data, including semi-structured and unstructured data, to build proactive supply chain optimization solutions.

- **Accelerate time to insight:** Unify SAP ERP data (inventory, orders, logistics) with external sources (weather, sensor, market, partner data) for a single, business-contextual view. Live data access to SAP enables detection of upstream delays, regional demand shifts and disruption risks before they escalate.
- **Scale with trust:** Build reliable, AI-powered applications based on a foundation of governed, semantically rich, business-context-aware data. Databricks Mosaic AI helps unlock critical use cases for supply chain optimization, activating agent systems against the most important supply chain data.
- **Future-proof supply chains:** Use Databricks AI and ML capabilities to forecast demand, optimize inventory, flag supplier risks and recommend replenishment or routing changes, enabling intelligent decision-making, automated process optimization and resilient operations.

Natura, a sustainability-driven Brazilian global personal care cosmetics group, leveraged SAP and Databricks to transform its complex supply chain and data landscape, integrating over 400 transactional systems — including Latin America's largest SAP installation — into a unified, analytics-ready platform. This Databricks and SAP solution enables automated, reliable and granular data, unlocking over 50 advanced analytics and AI initiatives. By optimizing data access and enabling deeper insights, Natura not only enhanced decision-making and efficiency across its vast supply chain but also amplified its positive societal impact, delivering R$2.7 in benefits for every R$1 of revenue.

# Next Steps to Data and AI-Supported Inventory

The transformative power of the Databricks Data Intelligence Platform leverages real-time data, scalable AI and seamless integration to enhance decision-making, optimize inventory management and ensure operational efficiency. The Databricks Platform empowers RCG, Manufacturing, Energy and HLS companies to stay agile, reduce costs and maintain a competitive edge in dynamic markets, revolutionizing supply chain management.

Learn more about how data and AI can improve operations while lowering TCO. Utilize the following resources and contact Databricks to explore the implementation and benefits of end-to-end inventory management.#

## Additional Resources

eBook

### Governance: The Unseen Foundation of AI Success

Develop a scalable AI strategy that provides built-in, customizable data governance protections. Databricks Unity Catalog gives organizations the tools to achieve data governance and AI goals within the Databricks Data Intelligence Platform. Secure data assets with a centralized approach, automatically monitor data and AI pipelines, and get unified visibility into data and AI.

Read now

Technical Guide

### Data Sharing and Collaboration With Delta Sharing

Data sharing is an essential component of driving business value, accelerating innovation and generating new revenue streams. Explore the power of data sharing using the Databricks Data Intelligence Platform, which helps the supply chain share data, analytics and AI across platforms, clouds and regions with strong security and governance.

Read now

eBook

### How to Transform Your Industry With Generative AI

Redefine what's possible with generative AI through more than 50 industry-specific use cases. See how the competition selects the best suppliers, manages with oversight, increases service delivery and reduces support costs. Apply these ideas, insights and anticipated outcomes within your enterprise to take advantage of scalable AI

Read now

## About Databricks

Databricks is the data and AI company. More than 10,000 organizations worldwide — including Block, Comcast, Condé Nast, Rivian, Shell and over 60% of the Fortune 500 — rely on the Databricks Data Intelligence Platform to take control of their data and put it to work with AI. Databricks is headquartered in San Francisco, with offices around the globe, and was founded by the original creators of Lakehouse, Apache Spark™, Delta Lake and MLflow. To learn more, follow Databricks on LinkedIn X and Facebook



© Databricks 2025. All rights reserved. Apache, Apache Spark, Spark and the Spark logo are trademarks of the Apache Software Foundation. Privacy Policy | Terms of Use